

# MazeX 1.0

## Technical Specification



**University of Moratuwa**  
IEEE Student Branch

**University of Moratuwa**  
IEEE Student Branch

A Student Chapter of the Robotics & Automation Society



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# **1. INTRODUCTION**

## **1.1. DESIGN. DEPLOY. DOMINATE.**

MazeX 1.0 brings together the sharpest undergraduate robotics minds to solve one of the most exciting autonomous robot challenges: the Micromouse maze. The mission is simple to state but difficult to master - design, build, and program a compact robot that can explore a maze, learn an efficient route, and reach the goal in the shortest possible time without human control.

Every turn, sensor reading, motor correction, and algorithmic decision matters. The best teams will combine embedded systems, mechanical design, control, and maze-solving intelligence to build a fast, reliable, and fully autonomous Micromouse.

## **1.2. About MazeX 1.0**

MazeX 1.0 is an intra-university Micromouse Robotics Competition organized by the IEEE Robotics and Automation Society (RAS) Student Branch Chapter and the IEEE Women in Engineering (WIE) Affinity Group of the University of Moratuwa.

The initiative is designed to promote innovation in robotics, algorithmic problem-solving, embedded systems development, practical engineering skills, and team collaboration. A pre-competition workshop series was conducted to introduce Micromouse concepts, maze-solving techniques, and robot design. These sessions provided participants with practical exposure and foundational knowledge prior to the competition.

## **1.3. The Micromouse Challenge**

The objective of the contest is to design and build a fully autonomous Micromouse capable of exploring and navigating a maze configuration and reaching the destination cell in the shortest possible time. The competition encourages students to apply classroom theory to real robot development and strengthens collaboration between RAS and WIE

## **1.4. Event Timeline**

Registration Opening : 7th July 2026

Registration Closing : 1st August 2026

Competition Day : 15th August 2026

## **2. THE COMPETITION**

### **2.1. Contest Eligibility**

#### **2.1.1. Team**

A team may consist of up to five members. All members of a team should be undergraduates from University of Moratuwa. Solo entries are also accepted.

Each team must nominate one representative for official communication.

#### **2.1.2. Registration**

All entrants must register before the registration closing date: 1st August 2026.

Each team should register under a unique team name.

If two or more teams request the same name, the name may be allocated on a first-come, first-served basis.

The organizing committee may reject inappropriate team names and request a replacement name.

The final registration procedure and official registration link will be announced through the official communication channels of MazeX 1.0 / IEEE Robotics and Automation Society (RAS) / IEEE Women in Engineering (WIE), University of Moratuwa.

## 2.2. Rules for the Micromouse

**Self-containment:** The Micromouse must be completely self-contained, fully autonomous, and must not receive outside assistance during a run.

**Power source:** The robot must not use combustion-based or pollutant-emitting power sources. The voltage between any two points of the circuit should not exceed 24 V at any time.

**Size:** The suggested maximum footprint is 14.5 cm length by 14.5 cm width. There is no restriction on height unless updated by the judging panel.

**Weight:** The weight of the Micromouse should remain constant during the competition. Battery replacements and approved repairs must not unfairly alter the robot weight or performance.

**Locomotion:** The Micromouse must not jump over, fly over, climb, scratch, cut, burn, mark, damage, or destroy maze walls.

**Dislodged parts:** The Micromouse must not leave any component behind while navigating the maze.

**External communication:** Wireless control or outside communication is not allowed during competition runs. Any RF or wireless module must be disabled or powered off during official attempts.

**Operation modes:** Manual switches may be used to select operating modes or algorithms. Reprogramming during an official trial or competition time is not allowed unless organizers explicitly permit it.

## **2.3. Rules for the Maze**

### **2.3.1. Maze Dimensions**

The proposed maze is composed of 18 cm x 18 cm unit cells arranged in a 16 x 16 grid. The walls are approximately 5 cm high and 1.2 cm thick, with reasonable construction tolerance allowed by the organizers.

The outside wall encloses the entire maze. The length from one wall side to the corresponding lattice point edge is approximately 19.2 cm. Final dimensions and tolerance values may be confirmed by the organizing committee before the competition.

### **2.3.2. Room Conditions**

The competition venue will use normal ambient room conditions. Teams should not assume a fixed level of temperature, humidity, sunlight, incident light, or fluorescent light at the venue.

### **2.3.3. Lattice Points**

Lattice points or posts are placed at the corners of unit cells. Teams should design their sensing and navigation logic to handle realistic construction tolerances and minor visual inconsistencies in maze materials.

### **2.3.4. Start and End**

The starting cell is located at one corner of the maze and is bounded by walls on three sides. The destination goal is located near the center of the maze. Multiple paths may exist to the destination, so teams should develop algorithms that can explore, map, and optimize rather than follow only one assumed route.

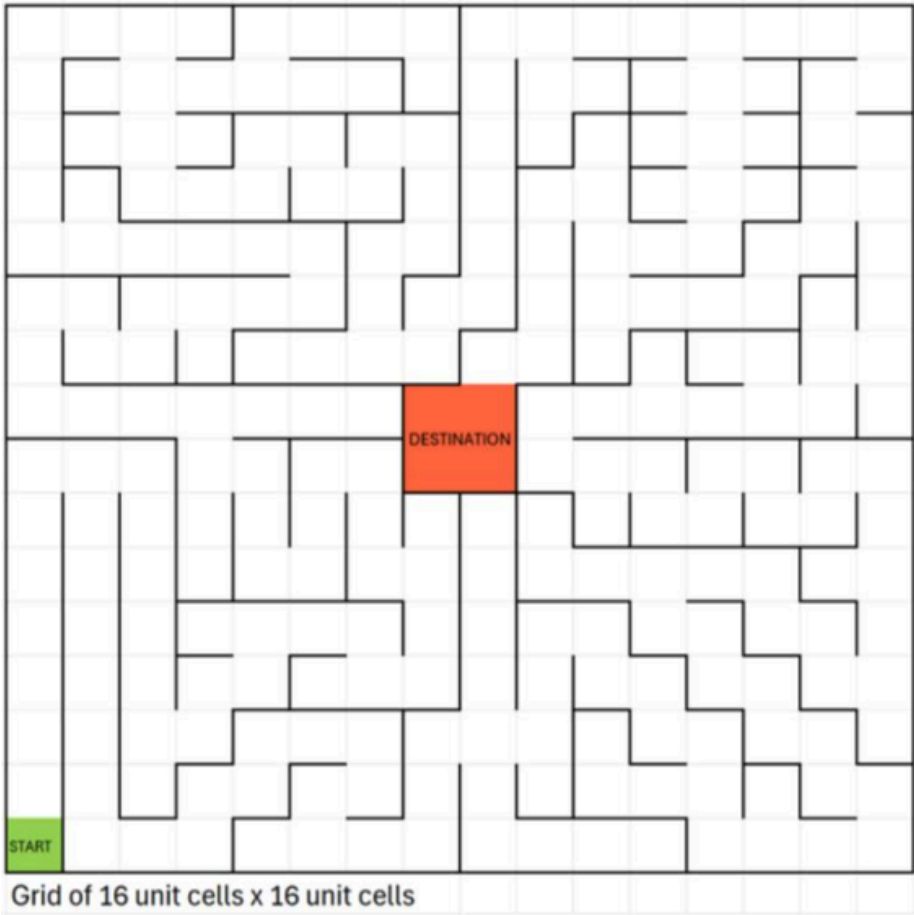


Figure 1: Sample Maze with Start and Destination

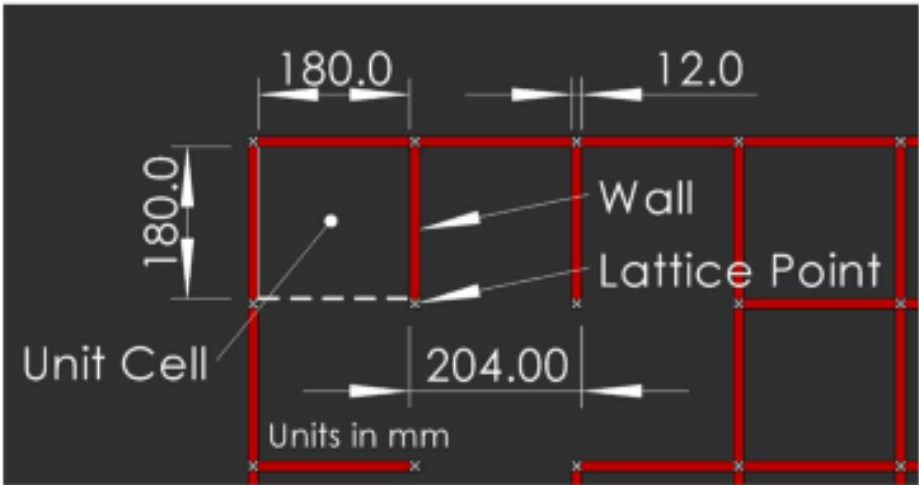


Figure 2: Maze Dimensions (units in mm)

## **2.4. Rules for the Elimination / Trial Round**

### **2.4.1. Initial Inspection**

Each team must pass an initial inspection by the judges to confirm that the Micromouse follows the technical and safety rules. Judges may request additional inspection at any time during the event.

### **2.4.2. Trial Time**

Each team may be given a fixed trial time allocation to access the maze. The timer starts when contest officials grant permission to begin. Any time taken for adjustments between runs is included in the allocated time.

### **2.4.3. Rounds**

A run begins when the Micromouse leaves the start cell and ends when it reaches the destination cell, returns to the start before reaching the destination, or is reset by the team. Teams may decide whether to perform search runs, fast runs, or a combination of both according to their strategy.



## **2.5. Rules for the Final Round**

### **2.5.1. Initial Inspection**

Each participating team shall pass the final inspection conducted by the judging panel to ensure conformity with the technical specifications defined under Section 2.2.

Following successful inspection, the Micromouse shall be placed in the designated holding area under the supervision of competition officials. The robot shall not be removed, modified, or reprogrammed without prior approval from the judges.

All teams must complete the inspection procedure before the official maze configuration is revealed.

Any robot failing to meet the required specifications or inspection deadline shall be disqualified from the Final Round.

### **2.5.2. Competition Time**

Each team is allocated 10 minutes access to the maze, referred to as the "Competition Time" and the timer starts when the contest officials acknowledge the contestant and grant permission to begin their attempt.

Any time taken to adjust the Micromouse between runs is also included in the 8-minute allocation.

### **2.5.3. Runs**

Each team will be given a maximum of 5 attempts or "Runs" during the allocated Competition Time. The decision for performing either "Search Runs" or "Fast Runs" is up to the discretion of the teams.

A Run is considered to have started once a Micromouse leaves the starting cell and ends when either the robot reaches the destination cell or returns to the starting cell before entering the destination cell.

Resetting the Micromouse back to the starting cell will also terminate the current Run.

#### **2.5.4. Run Time**

The run timer will start when the front edge of the Micromouse crosses the Start Line and stops when the front edge of the Micromouse crosses the Finish Line.

The time taken to travel from the start cell to the destination cell with any incurred time penalties is considered as the "Run Time" of the Micromouse for that attempt.

The Run Time will be measured by the contest officials and will be displayed during the competition.

The minimum Run Time within the Competition Time shall be the mouse's Official Time. First prize goes to the mouse with the shortest Official Time. Second prize to the next shortest, and so on.

### **2.5.5. Penalties**

#### **2.5.5.1. Contact Penalty**

The Micromouse will incur a 3 second penalty for each collision or contact made with the walls of the maze.

Successive collisions within a period of 3 seconds will be considered for a single penalty.

#### **2.5.5.2. Reset Penalty**

Once the Micromouse completes a run by travelling from the start cell to the destination cell, the run is considered successfully completed.

However, if the contestant opts to manually lift the Micromouse from the destination cell and place it back at the starting cell, this will incur an additional time penalty of 30 seconds to the current Run Time of the Micromouse.

### 2.5.6. Restarting the Micromouse

In the event that the Micromouse gets physically stuck within the maze and is unable to move, the contestant may request the judge's permission to Restart the mouse by manually lifting it and placing it back at the starting cell.

This would terminate the current Run and no Run Time would be recorded.

### 2.5.7. Adjusting the Micromouse

No information on the maze configuration can be entered into the Micromouse and software modifications are not allowed during the competition once the maze has been revealed. The Micromouse should be fully autonomous and can be switched on or off when inside the start cell.

Any adjustments to sensors / mode changes using the permitted manual switches can only be made once the Micromouse has ended the current Run. Any changes made during a Run would terminate the Run and a Run Time would not be recorded.

When the Micromouse is inside the start cell and has not started a Run, contestants are allowed to:

- Change switch settings (e.g. to select algorithms)
- Replace batteries
- Adjust sensors
- Change speed settings
- Make repairs (with approval from judges)

#### 2.5.7.1. Replacing Components and Making Repairs

Depleted batteries can be replaced with components of similar specifications without changing the overall weight of the Micromouse.

**Note:** Contestants are required to bring their own chargers / spare batteries and make sure that the robot is ready for operation during the allocated time slot.

Replacing any other components which may be damaged or worn out during the competition would require approval from the panel of judges.

However, the replacement parts should match the original component and should not alter the control characteristics of the Micromouse.

Replacement of microcontrollers or memory modules with permission from the judges would require the original software / code to be used and the contestant must assure the judges that no modifications to the code or addition of maze information has been made.

**Note:** Judges have the final say in allowing or denying any adjustments or replacements to the Micromouse by the contestants.

### **2.5.8. Additional Rules**

If a Micromouse is unable to complete a successful Run during the allocated Competition Time, the contestant will be given a score based on the proximity of the Micromouse to the destination cell and the total number of cells explored.

This score would only be considered if there were no contestants with Run Times eligible for the prizes.

The decision of the judges will be final.

## **3.Registration Fee Payment**

To complete your registration for MAZEX 1.0, please make the registration fee payment to the following bank account.

### **Bank Account Details**

- **Bank:Seylan Bank PLC (6287)**
- **Branch:Kalutara (Branch Code: 00036)**
- **Account Number:036034323059120**
- **Account Holder:SASINDU SIYATH WELLAGE**
- **Registration Fee : RS1500**

## Payment Instructions

- Please include your Team Name as the payment reference or description whenever possible.
- After completing the payment, upload the payment slip through the registration form.
- Registrations will only be considered complete after the payment has been verified.
- Keep a copy of the payment receipt until the competition concludes.

Note: If you experience any issues with the payment process, please contact the MAZEX 1.0 Organizing Committee before submitting your registration.

## **4. ADDITIONAL INFORMATION**

Any amendments to the rules will be updated through official MazeX 1.0 communication channels. Please do not hesitate to contact the organizing committee if further information or clarification is required. Participants are advised to regularly check official announcements regarding updates to competition rules, schedules, and technical specifications.